Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

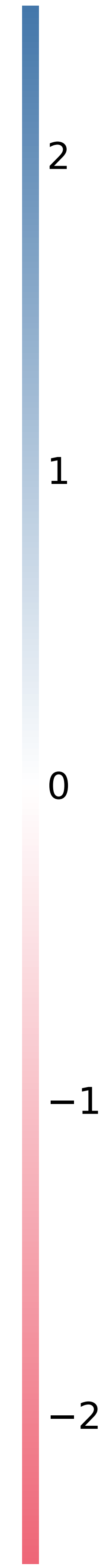
F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal



Reasoning + Answer:

<think>

To calculate the quantity of heat required to convert a liter of water at 30Â°C to a liter of water at 60Â°C, we first note that the difference in temperature is 30Â°C.

We multiply this with the amount of water (1 liter) and the specific heat of water, which is 1 cal/gÂ°C. Since 1 kg = 1000 grams, we multiply this by 1000. So the heat required is given by 1000 * 30 * 1 = 30 Kcal.

To calculate the quantity of heat required to heat a 1 kg block of aluminum from 30Â°C to 60Â°C, we first convert this temperature difference to Celsius degrees: 60Â°C - 30Â°C = 30Â°C

Since the density of aluminum is not given in this question we have to assume that it 1, so we multiply the block mass by the density * the specific heat, which are the same in this case. So the heat required is given by 1000 * 30 * .215 = 6.3 Kcal.

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<answer>G